

Search for Long-Lived Particles with HCAL Segmentation in CMS at the Large Hadron Collider

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Abstract

A search for long-lived particles (LLPs) with lifetimes greater than 0.1 ns is presented, using data from the Compact Muon Solenoid (CMS) experiment. Such particles are predicted by various theories beyond the Standard Model (BSM) that address phenomena like dark matter and supersymmetry. The search targets Higgs boson decays to a pair of massive LLPs (X), which subsequently decay to a final state with two bottom quark-antiquark pairs: $H \rightarrow XX \rightarrow b\bar{b}b\bar{b}$. The analysis utilizes the depth segmentation of the CMS Hadronic Calorimeter (HCAL) to identify the distinct topology of LLPs decaying within its volume. Two neural network (NN) classifiers are employed: a “depth” tagger, which uses information on the jet’s decay depth, and an “inclusive” tagger, which does not. This paper optimizes the signal region for an “asymmetric double-tag” strategy that requires at least one LLP to decay in the HCAL. Thresholds on the depth and inclusive scores are chosen to maximize statistical significance. We find that signal events for different LLP mass points populate distinct regions of the score space, generally favoring either high depth and low inclusive scores, or vice versa. Further investigation focuses on defining a unified optimal boundary.

1 Introduction

Long-Lived Particles (LLPs), often classified as particles with lifetimes greater than 10^{-10} seconds, feature in both the Standard Model (SM) – such as charged and neutral kaons and pions – as well as in many Beyond the Standard Model (BSM) theories, such as those relating to dark matter, the matter anti-matter asymmetry, and supersymmetry [1]. In this paper, we study the process, $H \rightarrow XX \rightarrow b\bar{b}b\bar{b}$, in which it is hypothesized that a Higgs boson first decays to an unknown long-lived massive particle (LLP), X , which in turns decays to the commonly observed final state consisting of pairs of bottom and anti-bottom quarks. The process is illustrated below in Figure 1. Our analysis will also provide insight into heavy twin Higgs boson models [2], since the masses of the Higgs boson and LLP are left as free parameters within our model. We may thus scan across various mass combinations, including heavier Higgs boson masses. The Higgs boson itself has been a focal point of study at the LHC ever since its discovery in 2012 [3], so the possibility of its coupling to an exotic particle, or the existence of its heavier twin, offers exciting potential for the discovery of New Physics.

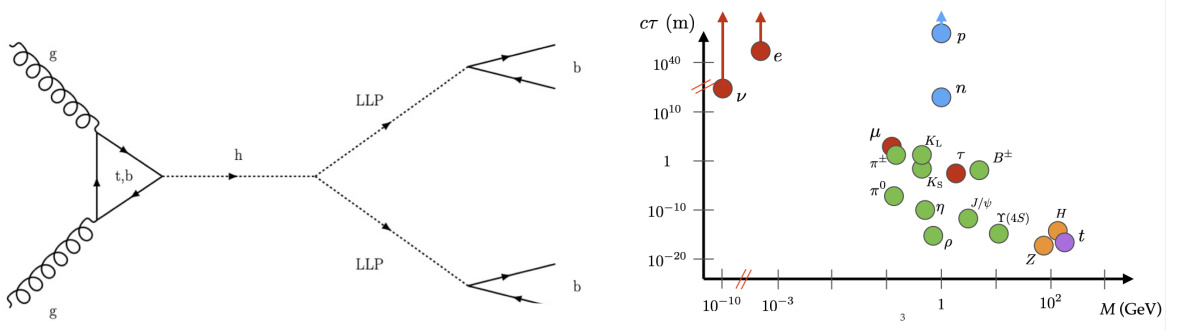


Figure 1: [Left] Feynman diagram showing the formation of a Higgs boson via gluon-gluon fusion and its subsequent hypothesized decay into a pair of LLPs which in turn decay to $b\bar{b}$ pairs. [Right] Plot of SM particles’ proper lifetimes (τ) in units of $c\tau$ as a function of their mass in GeV. [4]

Many of the particles that high energy physicists have been interested in decay almost instantaneously after they are produced, such as the Higgs boson itself with a lifetime on the order of 10^{-22} seconds. A short-lived particle would therefore decay effectively at the center of the proton-proton interaction point, subsequently producing many decay products such as leptons, photons, and hadrons. The subdetectors of the CMS experiment, shown in Figure 2, are optimized to detect the many signatures that the original particle leaves behind. For instance, the CMS silicon-based inner tracker records the trajectory of charged particles through the readout of electrical signals, while the electromagnetic calorimeter (ECAL) and hadronic calorimeter (HCAL) record the energies of electrons and photons, and hadronic particles, respectively. The information from these many sub-detectors is then used to reconstruct the masses, momenta, and charges of the decay products, from which the identity of the original particle can then be deduced.

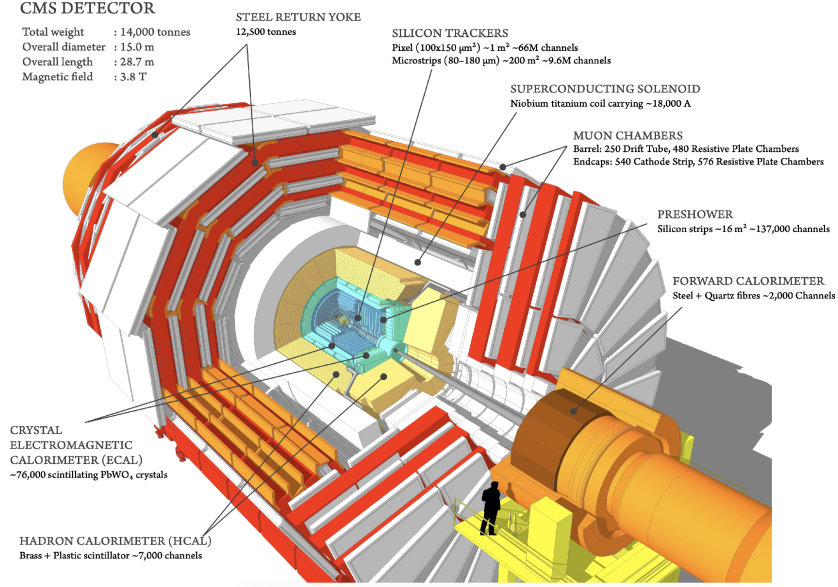


Figure 2: Diagram to show the components of the CMS detector [5].

By contrast, LLPs such as those hypothesized in the $H \rightarrow XX \rightarrow b\bar{b}b\bar{b}$ process could travel up to several meters before they decay, and thereby leave relatively fewer signatures within the detector, especially within its innermost parts such as the inner tracker or the ECAL. However, this absence of signal is a signature in and of itself. Indeed, this is the aim of the joint effort of the Caltech and Princeton groups in their analyses searching for LLPs, exploiting their distinct topology including a lack of signals in the inner detector regions.

One principle of our search strategy lies in selecting events with LLPs decaying within the HCAL because their signatures look very different compared to our main sources of background, such as QCD, Z + jets and W + jets, which form ‘prompt’ jets, as opposed to delayed ones. These differences are illustrated below in Figure 3. Firstly, we observe that a higher fraction of delayed jets’ energy would be deposited in the outermost layers of the HCAL, since we require that the LLP decays inside of it. We expect a more even distribution of the jet energy fractions from the prompt jets where a larger fraction tends to be deposited within the inner layers. This is a consequence of the fact that the HCAL is made of very dense material and so a jet produced before the HCAL is less likely to travel far inside of it and deposit its energy far into its depths. Similarly, due to the dense HCAL material, we also see that the jets produced from within the HCAL are less ‘spread out’ compared to jets produced immediately within the LHC beam, as a result of their displaced vertices.

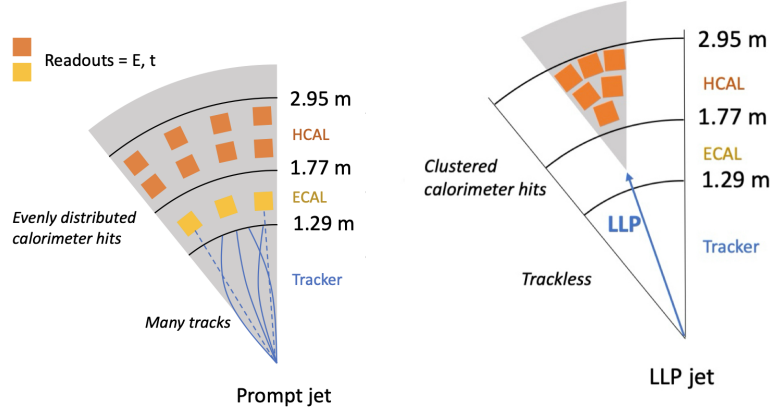


Figure 3: Illustration of a cross-section of the CMS detector to show the difference in the signature left by a prompt jet (left) and a delayed LLP jet (right). [4].

2 Background

The work presented in this paper builds on the offline analysis conducted during last year's SURF program which primarily involved the design and training of neural networks to distinguish between (Monte Carlo generated) signal events and background events, obtained from Run3 of the LHC, selected using a novel level 1 trigger, specifically designed for LLP selection by the Princeton group.

Due to the sheer number of collisions, occurring at a rate of billions of proton-proton collisions per second, it is not possible to store all the events in memory. Therefore, a series of triggers have been developed to determine the selection of events that are kept, and which are discarded. The Level 1 trigger is the first of such triggers, which processes events at a rate of 40MHz and reduces the output of information to 100kHz.

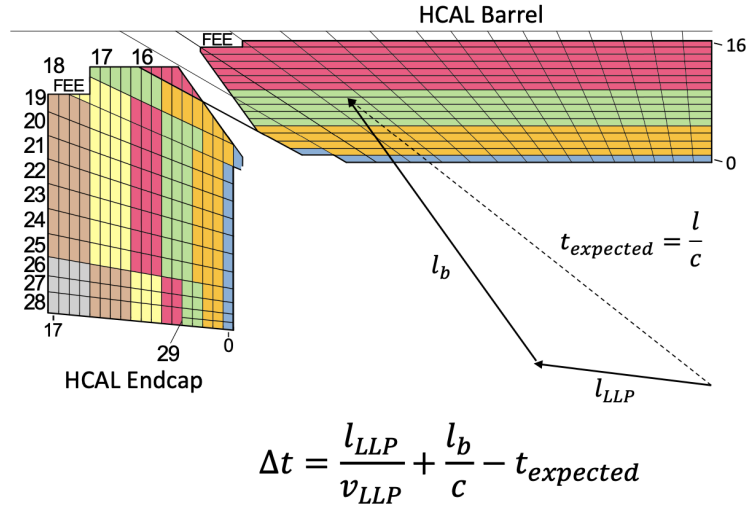


Figure 4: Diagram to show the resulting path difference due to the displaced vertex. Further, observe that the four colors denoting the calorimeter at the barrel represent the four layers of segmentation.

The LLP trigger is composed of several metrics. Firstly, it triggers on jets which deposit a higher energy fraction in the outermost layers of the HCAL. To measure this, one takes advantage of the recent detector upgrade which introduced depth segmentation in the HCAL. Namely, the HCAL has now been divided into four distinct layers, represented by the different colors on the HCAL barrel in Figure 4, where the trigger selects jets with high energy fractions in the third and fourth outermost layers.

Secondly, the trigger also measures the time delay between the expected arrival of a jet and its actual arrival. For jets produced by LLPs, we would expect to detect a time delay for two reasons. Firstly, there will be a time delay simply because the LLP will not decay immediately and so the jet will not be formed instantaneously either. Secondly, by the time the LLP will have decayed, it would have traveled a given distance from the center of the proton-proton interaction point. This means that if we were to reconstruct the trajectories of the particles in the jet, then their trajectories would not originate from the interaction region but would rather be formed at a displaced vertex. As a result, we see that this would create a path difference between a jet which simply traveled straight from the center to the point where energy is deposited in the HCAL versus if it were first originated from a displaced vertex. This path difference would thus contribute its own time delay. As a result, the LLP trigger is also designed to select events with at least one jet which arrived late. The trigger is designed to have three categories: prompt jets are labeled as events with a time delay of less than 6 ns, slightly delayed jets have a delay between 6-7 ns, while delayed jets are those with a delay of greater than 7 ns. The time of arrival of these jets is calculated relative to the HCAL clock which records the time relative to a particular bunch crossing. A custom data-taking run was performed to calibrate and align these time measurements.

Finally, the trigger has a minimum threshold jet energy of 40GeV and an implicit pseudorapidity, η cut of 1.4, in the sense that the trigger is designed to only detect events occurring in the barrel region of the detector as opposed to the end-cap, see Figure 4.

With the LLP Trigger in place, the aim of my work during last year’s SURF program was to optimize the offline analysis to further maximise our signal selection, especially at very high values of background rejection. The implementation of more advanced classification techniques was necessary to fulfill this goal due to the small cross-section of the Higgs boson decay at a mass of 125 GeV and at a total center of mass energy, $\sqrt{s} = 13.6\text{TeV}$, namely $59.2^{+5\%}_{-7\%}$ pb [6]. With an integrated luminosity of 160fb^{-1} reported so far for Run 3 in September 2024, [7], we expect to find on the order of just 10 million Higgs boson decay events across the entirety of the Run 3 dataset. Thus, within our dataset of 100 million events that passed our L1 trigger, we would expect to be left with even fewer signal events. Moreover, although the Higgs boson to LLP decays are not predicted by the standard model, previous combined analyses of the known Higgs decay channels have left the possibility that the ‘unknown’ decay branching ratio could be as high as 11%, [6].

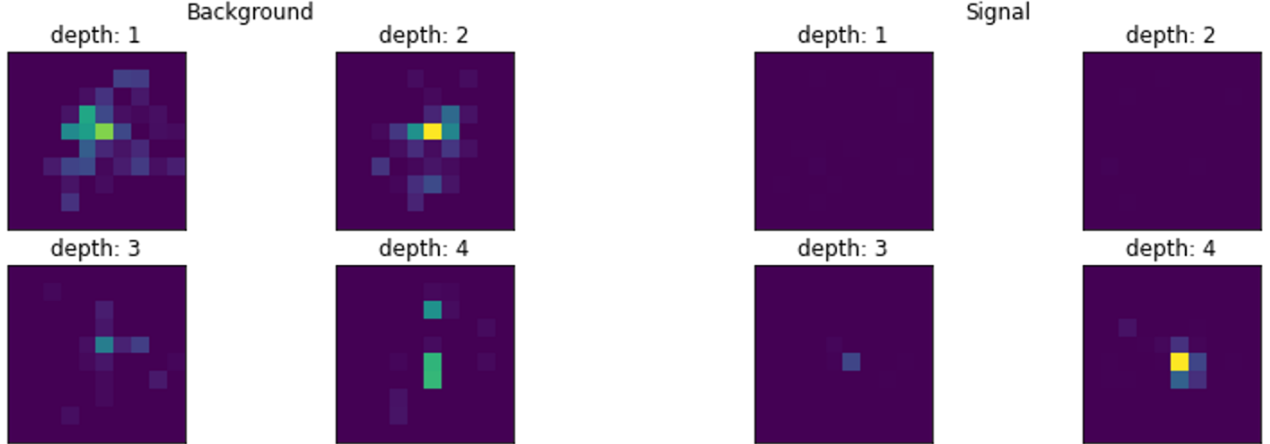


Figure 5: 3D jet ‘image’ samples of a background (left) and signal (right) jet. Each image is normalized to preserve the relative value of energy in each depth. These images align closely with what we would expect from our previous illustration of the differences of the LLP and prompt jets: the background has a more widespread distribution of energy across the depths of the HCAL, whereas the signal has a very localized energy deposit at the outermost layer of the HCAL.

Over the course of my project, I developed a number of classifier models with various architectures, such as a Dense Neural Network (DNN), a 3D Convolutional Neural Network (3DCNN) and even experimented with an autoencoder architecture and residual neural networks. Of these architectures, the DNN and 3DCNN proved to be the most successful as they were able to remain sensitive ($\approx 30\%$ signal efficiency) even down to a background efficiency of 0.01%. The DNN was trained on a whole host of features relating to the jets which include tracking information, jet energy, mass, and momentum, information about the fraction of the jet’s energy deposited in different layers of the HCAL, as well as custom variables such as $S_{\phi\phi} \equiv \frac{\sum \Delta\phi_i^2 \cdot E_i}{E_{\text{tot}}}$ which can be interpreted as a

measure of the ‘spread’ of a given jet. By contrast, the 3DCNN was trained on the ‘images’ (see Figure 5) of the energy deposited by the jets within the various layers of the HCAL. This model was trained on an ‘augmented’ image dataset, meaning that the model was trained on the original generated image samples, as well as their rotated and mirrored images. Given that the signature of an LLP is intimately tied to the topology of the jet produced when it decays within the HCAL, it would be more natural to represent jets in a way that preserves this spatial information about their structure rather than simply storing this information in one-dimensional arrays of jet metrics, as in the case of the data inputted to the DNN. This reasoning can help us to explain the better overall performance of the 3DCNN in comparison to the DNN when evaluated based on their Receiver-Operating Characteristic (ROC) curves, presented in Figures 6 and 7.

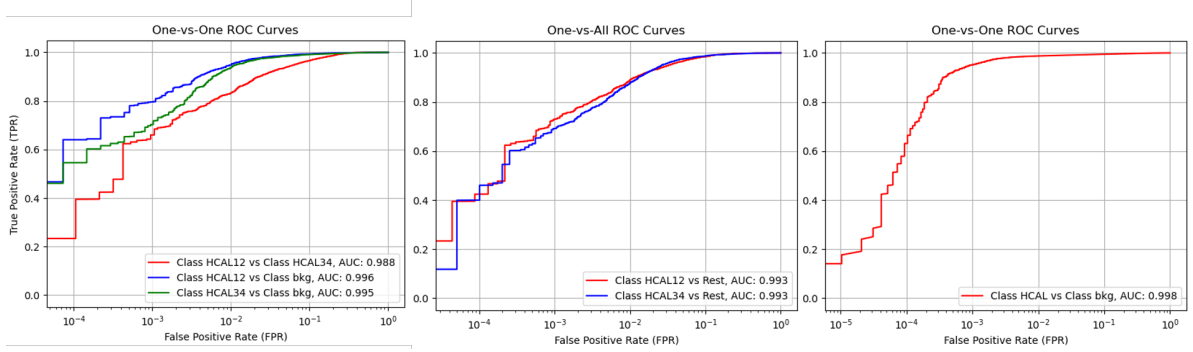


Figure 6: Series of ROC curves displaying the performance of Dense Neural Network implemented in Tensorflow’s Keras and trained on the MH350-MS160 dataset. [Left] ROC curves shown for how well the model distinguishes between the two classes. [Center] ROC curves showing how well the model distinguishes one signal class from the rest. [Right] ROC curve showing the performance of the model as a binary classifier, where the two signal classes are combined into one.

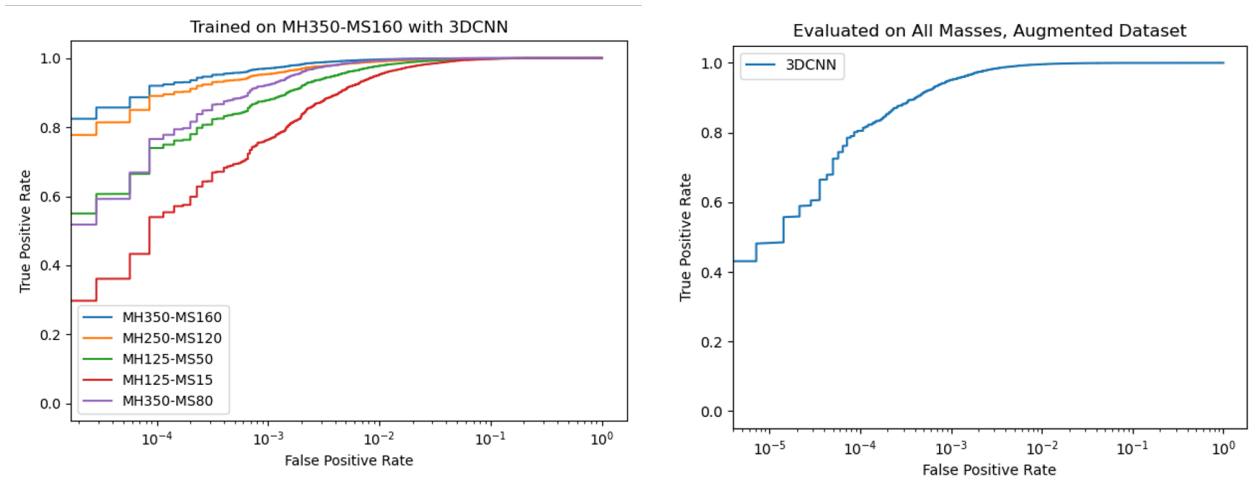


Figure 7: ROC curves showing the performance of the 3DCNN model. [Left] Comparison of the performance of the 3DCNN for different values of the BSM Higgs boson mass MH and the scalar mass MS. [Right] Combined mass evaluation with augmentation of dataset (rotated and flipped image inputs).

3 Methodology

This time last year, my project culminated in the production of several powerful LLP classification models which had been found to improve background rejection by at least an order of magnitude. Nevertheless, these models only existed in ‘isolation’: they had not yet been integrated into the official analysis workflow or applied as part of a formal search strategy. At present, however, these models have now evolved into forming the foundation for

the ‘asymmetric double-tag’ search strategy, which is outlined as follows. In this approach, we use two separate deep neural networks (DNNs) to analyze the two LLP jets produced within a single Higgs boson decay event (as illustrated in the Feynman diagram in Figure 1). Firstly, we select events which have at least one jet decaying within the HCAL. This offers a good balance between rejecting noise yet still retaining a large number of events in our dataset. The jet that decays within the HCAL is then passed through the “depth” tagger, which is trained on information including the fraction of energy deposited in each of the layers of the HCAL. The output of this depth classifier for a given jet is then a score, where jets with a score above a predefined threshold are deemed to result from LLP decays. Similarly, the jet that is free to decay anywhere within the detector is passed to an ‘inclusive’ tagger. Then, we can represent our events within a 2D space, with each event consisting of a pair of jets assigned two numbers: its inclusive score and its depth score, as illustrated in the Figure 8 below.

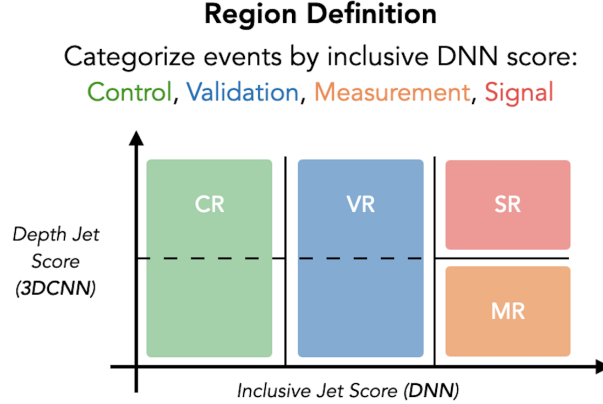


Figure 8: A visualization of the plot of the inclusive tagger score versus the depth tagger score. The dashed horizontal line represents a chosen fixed score threshold above which we treat events as signal, and below which, as background. The regions shown, which are explained in the text are CR: Control Regions. VR: Validation Region. MR: Measurement Region. SR: Signal Region.

In the simplest case, we assume that the depth and inclusive scores are uncorrelated which corresponds to the case where the jets behave as independent objects. This then implies that within any specific bin of DNN scores, the fraction of events classified as signal or background by the depth tagger remains constant. We can validate this assumption by examining the control (CR) and validation (VR) regions (see the figure above): we check for the proportion of signal and background (based on the depth score threshold). If the proportions within the CR and VR bins are consistent, we can extrapolate this consistency into the signal region (SR), which allows us to predict how many events should appear there, based on the number of events within the measurement regions (MR). Comparing the actual number of events in the SR to this prediction then informs us as to whether there is a significant discrepancy that might indicate new physics. The goal of my SURF project is therefore to find an optimum set of thresholds that maximize the statistical significance of our counting experiment, by aiming to maximize the signal selection efficiency and the background rejection.

To achieve this, I have designed a workflow spanning several scripts to predict the number of events in each bin based on a threshold (or in the case of signal files, to calculate the number directly), compute the statistical significance based on the predictions, and visualize the results as a scan across the threshold space. This is set to $(0, 1) \times (0.5, 1)$ where the former and latter set in the cartesian product correspond to the depth and inclusive scores respectively. It should also be noted that this is a “blind” analysis, where the data in the signal region is blinded until the selection criteria are set, and all validation steps and systematic checks are done, using the data outside of the signal region. The workflow is outlined below.

Initializing the regions: Defining histogram cuts based on the DNN depth and inclusive scores, using the desired/ inputted set of thresholds.

Applying physical cuts, weights, and binning data: We apply physical cuts by requiring that the two jets in the event are tagged as an LLP candidate by the L1 trigger (“DepthTagCand” and “InclTagCand”). These cuts reduce the size of our background datasets by approximately two orders of magnitude, leaving on the order of 10^5 events. The remaining events are then binned according to their transverse momentum,

eta, and phi metrics, each partitioned by 10 bins, resulting in 10^3 bins in total. For all signal files and calculations, a weight is applied to the histograms that corresponds to the physical branching ratio of the process $H \rightarrow XX \rightarrow b\bar{b}b\bar{b}$. This weight is derived from estimates and Monte Carlo simulations. As a result, we also vary these weights by factors of 0.05, 0.1, 1, and 5, to gauge how the resulting significance varies.

Calculation of Mistag Rate: The mistag rate is calculated for each bin in the Control Region: it is the fraction of events in the bin that are ‘mistagged’, divided by the total number of events in the bin. The mistag rate is 0 for any empty bin.

Validating the Mistag Rate: Using the calculated per-bin mistag rate, we perform per-bin multiplication to calculate the expected number of events in each bin. This prediction is compared with the true data: both in terms of total number of predicted events, and per-bin comparisons. The total predicted number of events for the 2023 Bv2 LLPSkim dataset (v3.13) agrees with the true number to within less than 1%. Figure 9 below shows a comparison of the predicted and true mistag rates for the transverse momentum bins.

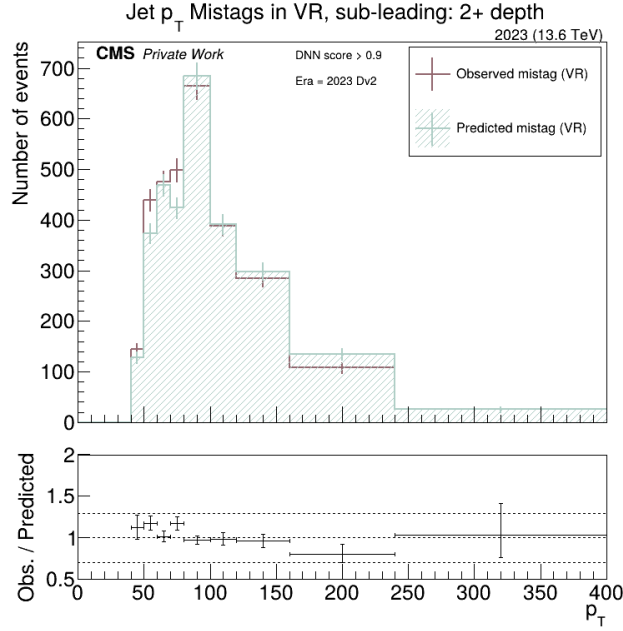


Figure 9: A plot showing a comparison between the predicted number of mistagged events in the Validation Region and the true number of mistagged events. The plot shows that the prediction captures the distribution of mistagged events well, though there are some discrepancies.

Predicting the SR region: In the same way that we calculate the prediction for the Validation Region, we predict the number of events in the Signal Region based on the total number of events in the Measurement Region. However, we cannot check the true number of events in the Signal Region prior to the official un-blinding of our analysis.

Aggregating the Data: After repeating the computation for each combination of thresholds in the space (sampled evenly by some spacing such as dividing the sets by 30), this data is saved. In a separate script, we aggregate data corresponding to a mass point and the background, where both pieces of information are essential for computing the rejection power/ statistical significance.

Computing the Significance: Now that the data is neatly aggregated, it is straightforward to compute the significance at every point in the parameter space of the depth score and inclusive score. The formula used for the statistical significance is given by

$$Z = \left(-\frac{1}{2(B \log(1 + \frac{S}{B}) - S)} + \frac{B}{S^2} + \frac{1}{S} \right)^{-\frac{1}{2}}$$

where S and B are the estimated counts of signal and background events that pass a applied threshold, respectively. This formula is an approximation for when $S \ll B$ of the Asimov significance for a single Poisson counting experiment [8].

Visualising the data: A heatmap is generated from the significance scans across the threshold space. Such heatmaps can also be generated to just show how the number of events varies in each signal file. Example plots of the scans of the number of signal events, the predicted number of background events, and the resulting significance scan (in units of standard deviations) for the case of a BSM Higgs boson mass of 350 GeV and X mass of 160 GeV, are shown below in Figure 10. A comprehensive webpage containing all of the plots with various configurations can be found at [9].

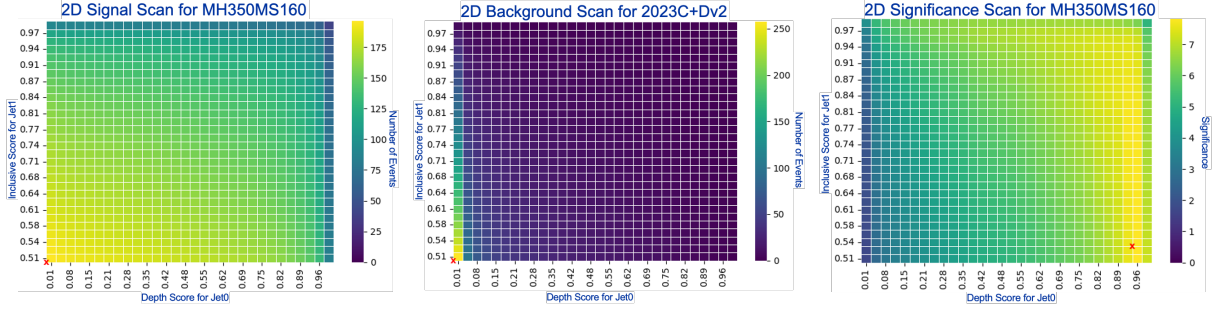


Figure 10: All three plots correspond to computations made with the following parameters in place: the depth score is calculated from the “anywhere” depth model. The inclusive score is calculated from the “train80” model². The event space is defined such that the leading jet (Jet0) decays within the HCAL and is passed to the depth model. The subleading jet (Jet 1) is passed to the inclusive model. Signal weights are applied, meaning that the numbers of events in the signal files are scaled to the number of events we would expect to see within our data, given an assumed branching ratio. The background is scaled by 25 times. A red ‘X’ marks the cell with the highest value. [Left] The calculated number of signal events remaining that pass the inclusive and depth score thresholds. [Center] The estimated number of background events that pass the threshold selections. [Right] The calculated significance derived from the two previous plots.

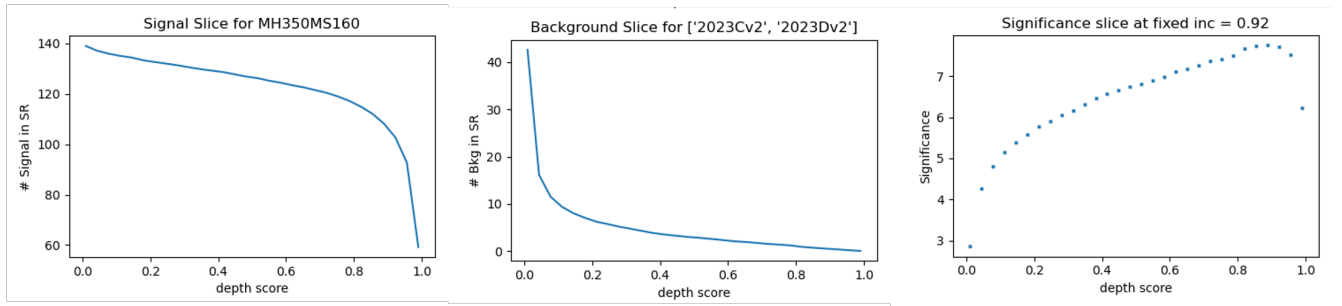


Figure 11: One-dimensional cross-sections of the 2D scans in Figure 10 above, defined by inclusive score = 0.92, for the mass point MH350-MS160. [Left] The variation of the number of signal events in the Signal Region as a function of the depth score. [Center] The variation of the number of background events in the Signal Region as a function of the depth score. [Right] The variation of the calculated significance (derived from the number of signal and background events in the Signal Region) in standard deviations as a function of the depth score.

To aid in interpreting these heatmaps, we also present a one-dimensional ‘slice’ of these 2D scans above which correspond to the horizontal ‘cross-section’ of the heatmaps defined by ‘inclusive score = 0.92’ (qualitatively, the horizontal variation near the top of the plots above). For the signal scan, we can see that there is a gradual decline in the number of signal events at lower depth score, before a much steeper decline occurs as the depth score approaches to 1. On the other hand, the background drops off much more steeply at lower depth scores.

² “Train80” refers to the inclusive model that is trained on 80% of the available data and validated with the remaining 20%. This model has shown the optimal performance out of other tested combinations, such as a train-validation split of 40%-60%.

These distributions make sense, given that the majority of signal events should be assigned high depth scores, while background events should be associated with low depth scores. The resulting significance distribution is not monotonic, as we can see in the rightmost plot in Figure 11. Although this may not necessarily be the case for every process, we can justify the non-monotonicity by thinking of the extremes: at a very low depth score, we have many background events, and hence the signal to background ratio is expected to be lower. On the other hand, at very high depth scores, we have cut out most of our background, but also at the cost of cutting out most of our signal. Hence, we argue that there must exist some intermediate compromise between these two extremes. Indeed, in the case of the MH350MS160 mass-point, the depth score that maximizes the significance at a fixed inclusive cut = 0.92, is depth score = 0.9. This is also roughly the point at which we observe the start of the steep decline in number of signal events.

Another relevant metric that we could consider is constructing a one-dimensional ROC curve for a given cross-section of the threshold space. This is done by fixing one of the depth or inclusive scores at a certain value (e.g. inclusive score = 0.9) and calculating the true positive rate (TPR) and false positive rate (FPR) based on the score that is allowed to vary. Note that the TPR is defined as the ratio of the passing signal to the total signal in the space (the maximum number of events in the slice). The FPR is defined similarly except it is the ratio of the remaining background to the total background. Below, we present the constructed family of ROC curves for MH350MS160. We can see that the family of curves is effectively invariant with respect to the variation of the fixed threshold. This therefore further helps to confirm the assumption that the depth and inclusive scores are independent given that the distribution of scores remains the same within any slice. Furthermore, from the shape of the curve, we see that the depth model is far more discriminating than the inclusive model, as we would expect from its access to the additional depth information that it is trained on. Note that the y-scales of the plots start from about 0.45, showing that there is still a significant signal efficiency at background efficiency $\ll 0.1$.

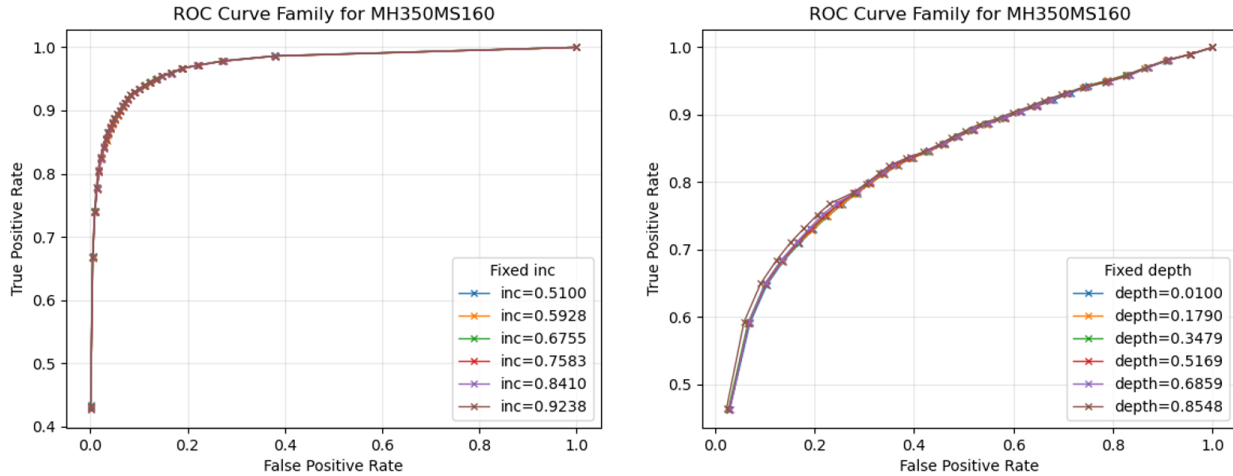


Figure 12: ROC curves constructed from a cross-section of the threshold space. [Left] ROC curve constructed from the depth model using the depth scores, obtained by fixing the inclusive score at various values shown in the legend, and calculating the TPR and FPR across the range of the depth scores. [Right] ROC curve family constructed for the inclusive model by fixing the depth score.

4 Results

Many significance studies were performed with a number of varying parameter configurations which include the mass points (e.g. MH350MS160), the passing of the leading jet to the depth model versus the passing of the sub-leading jet to the depth model, and the weights of the events. Several observations based on these variations are discussed below.

4.1 Varying Mass Points

From Figure 13 below, we can see that the maximal significance, marked by the red 'X', falls consistently within the same horizontal region, corresponding to 0.89-0.95 depth score. Although there does appear to be a large spread

in the vertical position of the maximal significance, it is important to keep in mind that, generally, the vertical variations in inclusive score are quite negligible for a fixed depth score. Hence, from these plots we conclude that the thresholds on the depth score must be far tighter compared to the inclusive score. This pattern could be explained by the fact that while increasing the depth score will roughly cut the background and signal by the same scale factor, the depth score is much more effective in discriminating against background, allowing it to preserve relatively more signal.

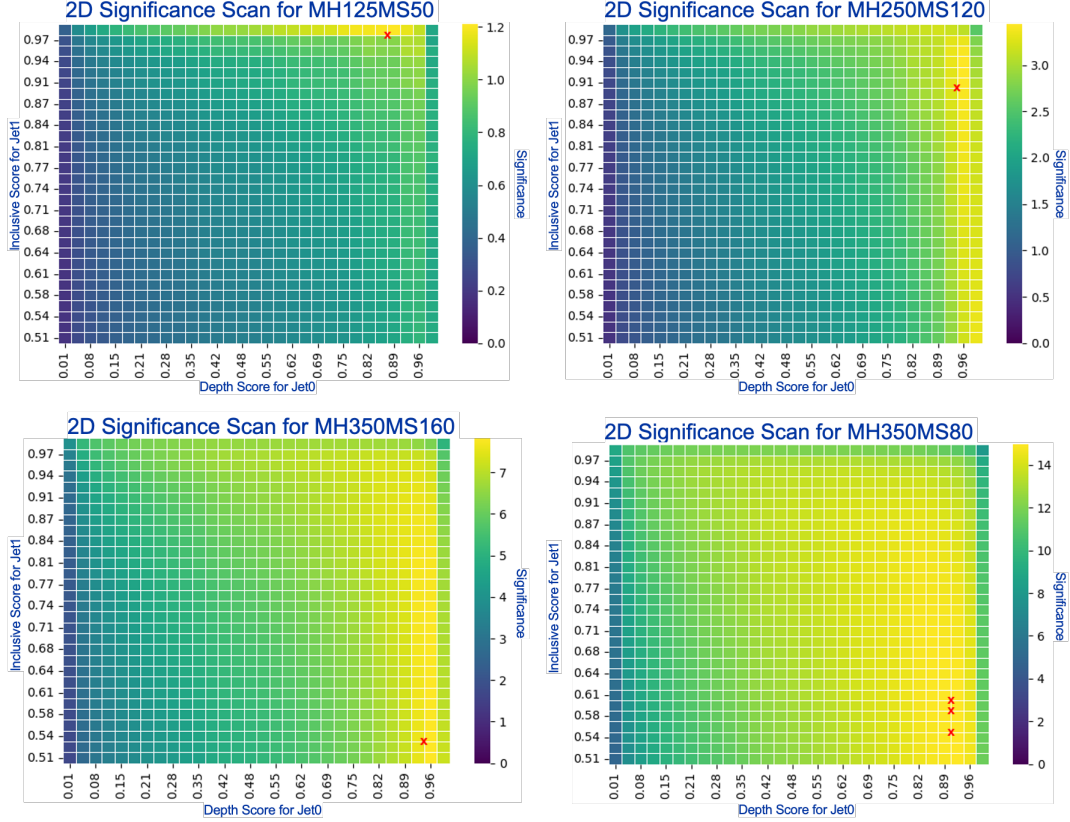


Figure 13: Shown above are significance scans for four mass points $(MH, MX) = (125, 50), (250, 120), (350, 160)$ and $(350, 80)$ GeV. The plots were generated with the following common configurations: the depth model is “anywhere”, the inclusive model is “train80”, and the depth tagger is applied to Jet 0.

4.2 Depth Tagging Leading vs Subleading Jet

The trends are less pronounced when varying this configuration across the mass points. However, one observation that can be made for several mass points such as the MH350MS80 and MH125MS50 sample is that, in going from leading to subleading jet, the maximal significance requires a much lower depth threshold. This would make sense given that there are fewer events which contain a subleading jet decaying in the HCAL, and hence to retain more signal events, the depth threshold should be loosened. On the other hand, it appears that loosening this threshold can be compensated by tightening the inclusive score, as is evident in the case of MH350MS80. We have also observed that across all four mass points, the maximal significance halves going from selection on the leading jet to the subleading jet, as illustrated in the plots in Figure 14.

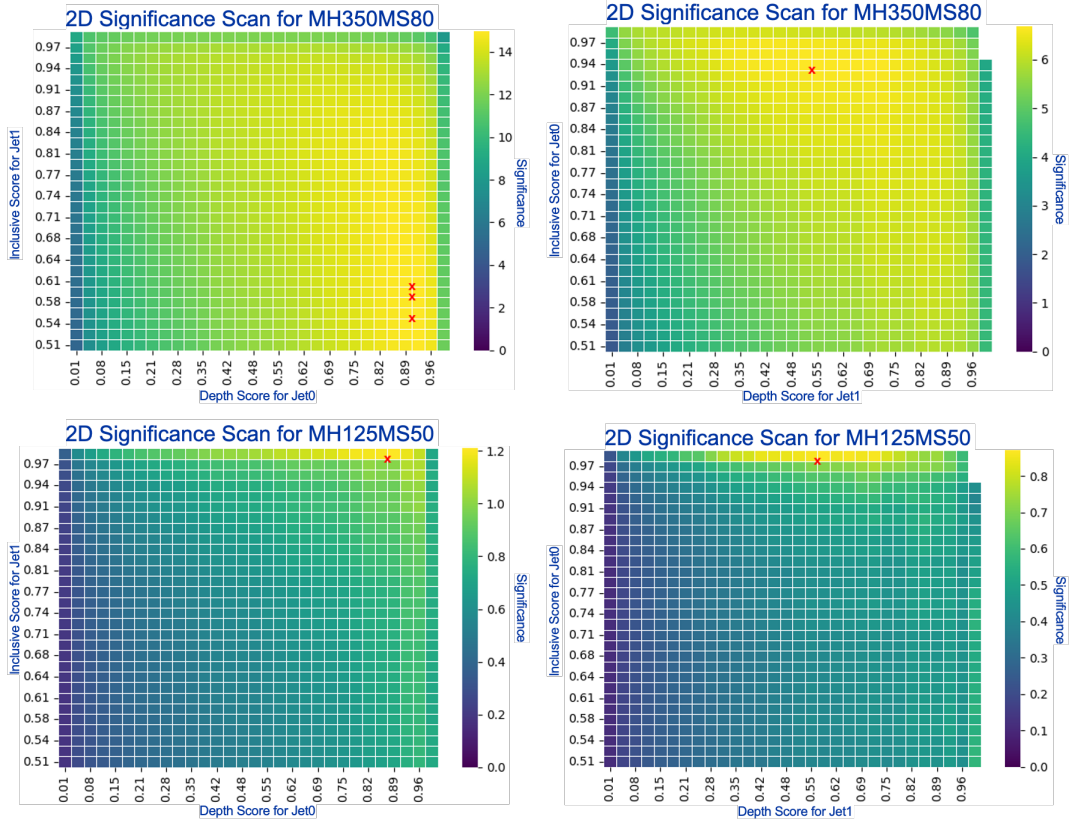


Figure 14: Shown above are the significance plots for two distinct mass points, corresponding to the two rows. In each row, the first column shows the scan for the leading jet depth tagged; the second column shows the scan for the subleading jet depth tagged. Note the differences in axis labels. The plots were generated with the following common configurations: the depth model is “anywhere”, while the inclusive model is “train80”.

4.3 Varying Event Weights

Event weights are varied by applying a scale-factor, w , to the computed signal weight from the Monte Carlo simulations. The branching ratio for each process is assumed to be approximately 1% so the applied event-weight is approximately $0.01 \cdot w$. Looking at a single mass point, MH350MS160, in Figure 15 below, we can see that while the overall significance tends to increase across the scan, it does not behave linearly with the applied scale-factor. It is interesting to see that the overall distributions for $w=0.05$ and $w=0.1$ are more similar to one another, relative to the comparison between $w=1$ and $w=5$. It is therefore of interest to explore intermediate values between 0.1 and 1 to explore how the distributions shift from one regime to the other.

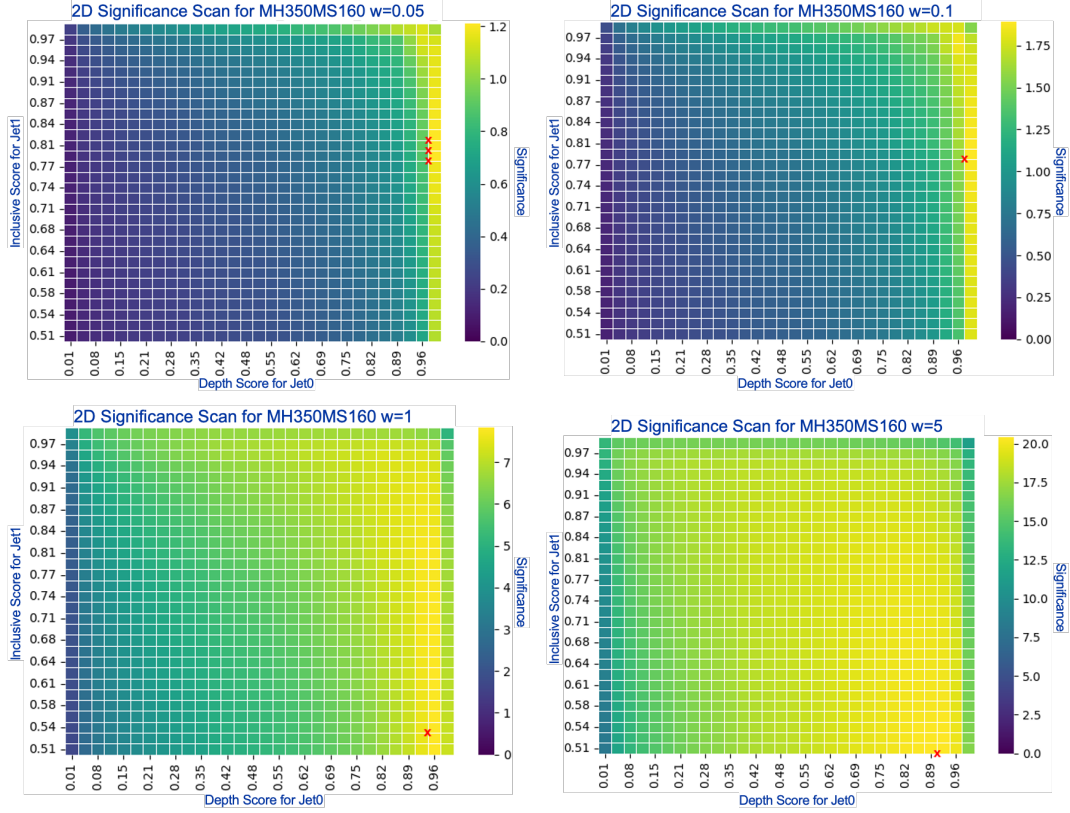


Figure 15: Shown above are significance plots for the mass point MH350MS160 for various values of the scale-factor w . The plots were generated with the following common configurations: the depth model is “anywhere”, the inclusive model is “train80”, with Jet 0 (the leading jet) passed to the depth model.

4.4 Exclusion Boundaries

We require a significance minimum of approximately 2 standard deviations to exclude a process. Therefore, we applied a mask to the regions of the threshold space where the significance exceeds 2 standard deviations. This enables us to visualize a clear exclusion boundary, as shown in the example plot below for MH250MS120. It was observed that the area of the exclusion boundary increases monotonically with higher Higgs and Scalar Masses. This makes sense given that our selection cuts require a transverse momentum threshold. Therefore, we retain a greater proportion of the higher mass signals as they are expected to have a higher transverse momentum on average, meaning that even at tight thresholds, there is sufficient signal among the background events to result in a significance above the 2σ threshold.

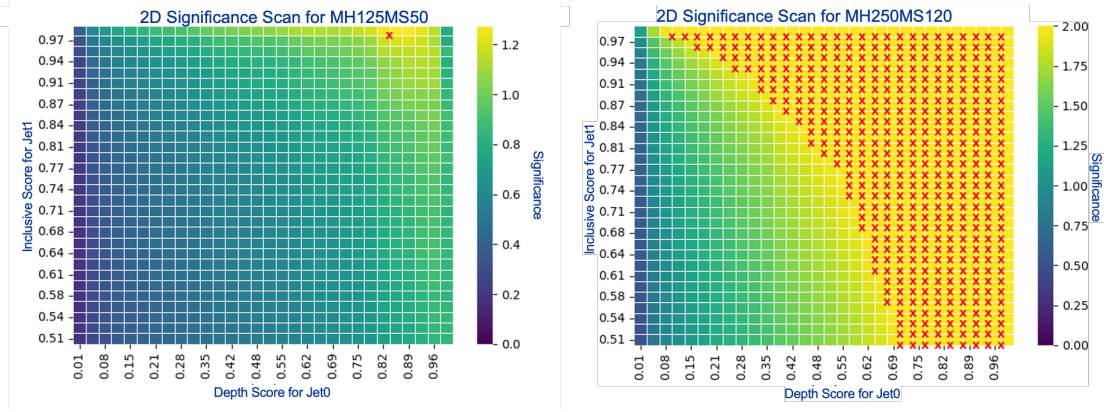


Figure 16: Significance scan for MH125MS50 and MH250MS120, with values clipped at 2. Significance values for MH350MS80 and MH350MS160 exceed 2 in all regions.

To help us choose a (inclusive score, depth score) pair that can exclude as many possible processes, we can extend the idea of an exclusion boundary to a *unified* exclusion boundary, which is obtained from the union of the $> 2\sigma$ masks generated for each mass-point at a fixed weight-fraction. To mitigate a null result, which will occur whenever at least one of the mass-points do not have any region where the significance is greater than 2 (an intersection with an empty set is itself an empty set), we first check whether all the mass-points actually possess a region of significance $> 2\sigma$; if not, these non-passing mass-points are simply dropped from the computation and we go on to generate the unified exclusion boundary for the rest of the non-empty masks. Thus, even when some mass-points are dropped, the boundaries generated can still be interpreted as the best possible set of thresholds which maximize the number of processes excluded. This choice to manually drop some mass-points then explains why we may see a ‘looser’ boundary generated for a smaller weight fraction, as seen in Figure 17 below. For a weight-fraction of 1, it is not possible to obtain a significance greater than 2 for the mass-point MH125MS50. Thus, in generating the left-hand plot, the file was dropped and hence the boundary corresponds to the space that can exclude the rest of the three out of the four total mass points. On the other hand, for a weight-fraction of 2.5, all four of the mass points had regions where significance exceeded 2σ . Thus any score combination within the boundary in the left plot will exclude all mass-points at a weight-fraction of 2.5.

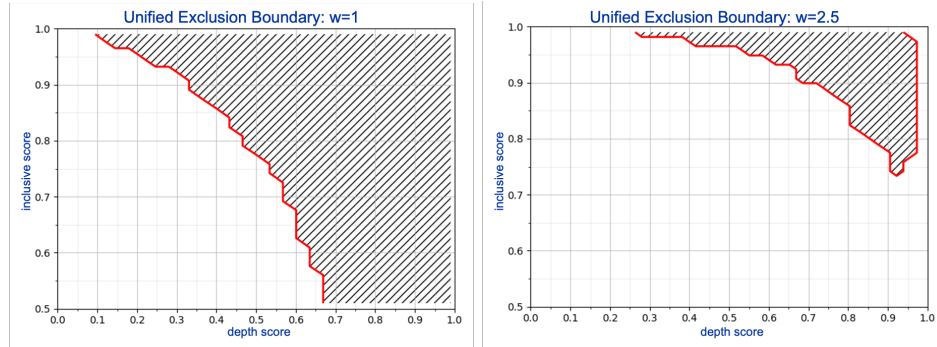


Figure 17: Plots of the unified exclusion boundary for fixed weight-fractions $w=1$ and $w=2.5$. In both plots, the region connected to the upper right area of the plot is the exclusion region.

4.5 Minimum Branching Ratio Sensitivity

Up until now, we have been working within a framework where we have a fixed set of processes defined by a discrete set of given mass-points and weight fractions with the aim of searching for regions in their threshold space with sufficient significance. However, we now turn to explore an alternative framework in which we perceive the weight fraction as continuous and instead we search for the minimum weight fraction for which we remain sensitive at greater than 2 sigma exclusion significance. In this new regime, we can rephrase our central question from “for a fixed branching ratio, what are the optimal thresholds to adopt?” to “given a fixed set of thresholds, what is the

minimum branching ratio for a given process that we are sensitive to?”. The minimum weight-fractions were found by performing a binary search, and are recorded in Table 1 to one significant figure. However, the computation of the minimum weight sensitivities for an exact minimum threshold yet remain to be implemented as the current workflow would not support this efficiently. From the data, we again see the monotonic behavior of the sensitivity in correspondence with the Higgs mass: the higher mass Higgs boson exhibit a smaller minimum weight-fraction sensitivity due to more events passing the p_T threshold. We also find that the sensitivity to the Jet1 category is lower than the Jet0 category. This also makes sense given that there are fewer such Jet1 events, and furthermore, that the sub-leading jet is likely to be lower in transverse momentum and hence more events are cut out in the selections.

Mass Point	J0 Minimum wfrac	J1 Minimum wfrac
MH350-MS80	0.04	0.2
MH350-MS160	0.2	0.4
MH250-MS120	0.4	1.6
MH125-MS50	2	3.8

Table 1: A table showing the minimum weight-fraction for a given process for which it is possible to obtain significance of $\geq 2\sigma$ for some combination of depth and inclusive score thresholds. The processes are split by the mass point and the jet that is depth-tagged (i.e. J0 corresponds to the data when the depth score pertains to the leading jet).

5 Conclusion

The work presented here offers a promising exploration of signal region optimization for the search for LLPs. It plays an important role in advancing our analysis, from simply the development of LLP taggers, to their actual application in defining a strategy to claim and justify statistically significant results. Our investigation has achieved the goal of creating the necessary computational methods to be able to efficiently calculate the significance across the model parameter space and the space of depth and inclusive thresholds, and produce useful visualizations to help inform our final threshold choices. From the analysis presented in this paper, we have drawn several conclusions regarding the trends of the distributions of the high significance regions across mass-points, weight-fraction, and tagging regime, thanks to the sufficient flexibility of our workflow which accommodates aggregation of data across various metrics. We have also explored the signal optimization from two frameworks which switch between treating the threshold space and branching ratio as discrete. While we have made significant progress towards narrowing down potential regions of optimal threshold selections, the determination of an exact threshold combination still remains to be seen and is a particularly promising extension of our current work.

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